

The Differential Analogue of Favard's Theorem

Statement of the Classical Theorem

Favard's theorem characterizes sequences of univariate functions $P = \{p_n\}_{n \in \mathbb{Z}_+}$ satisfying the three-term multiplicative recurrence

$$\begin{aligned} p_0(x) &= 1, & xp_0(x) &= \beta_0 p_0(x) + \gamma_0 p_1(x), \\ xp_n(x) &= \alpha_n p_{n-1}(x) + \beta_n p_n(x) + \gamma_n p_{n+1}(x), & n &\geq 1, \end{aligned} \quad (\text{F})$$

for real sequences $\{\alpha_n\}_{n=1}^\infty, \{\beta_n\}_{n \in \mathbb{Z}_+}, \{\gamma_n\}_{n \in \mathbb{Z}_+}$. The theorem states: (F) holds if and only if p_n is a polynomial of degree n for all $n \geq 0$ and there exists a finite, signed Borel measure μ on $\mathbb{R}$ with

$$\int_{-\infty}^{\infty} d\mu(x) = 1, \quad \int_{-\infty}^{\infty} p_m(x)p_n(x) d\mu(x) = 0 \quad \text{for all } n \neq m.$$

Three ancillary facts:

1. μ is a positive measure if and only if $\alpha_n \gamma_{n-1} > 0$ for all n .
2. μ is symmetric, i.e. $d\mu(-x) = d\mu(x)$, if and only if $\beta_n = 0$ for all n .
3. P are orthonormal with respect to a symmetric μ if and only if $\alpha_n = \gamma_{n-1}$ for $n \geq 1$.

The measure μ is unique only up to diagonal rescaling of P ; it may also be non-unique if the moment problem $M_n = \int_{-\infty}^{\infty} x^n d\mu(x)$ is indeterminate.

Statement of the Differential Analogue

The differential analogue replaces the multiplicative operator $x \cdot$ with the differential operator $\frac{d}{dx}$. The object of study is a sequence $\Phi = \{\varphi_n\}_{n \in \mathbb{Z}_+}$ satisfying

$$\begin{aligned} \varphi'_0(x) &= \beta_0 \varphi_0(x) + \gamma_0 \varphi_1(x), \\ \varphi'_n(x) &= \alpha_n \varphi_{n-1}(x) + \beta_n \varphi_n(x) + \gamma_n \varphi_{n+1}(x), & n &\geq 1, \end{aligned} \quad (\text{D})$$

for complex sequences $\{\alpha_n\}, \{\beta_n\}, \{\gamma_n\}$, where φ_0 is a given smooth function (and $\varphi_{-1} \equiv 0$ by convention). The paper of Iserles and Webb (2020) characterizes when such systems exist, when they are orthogonal, and when they are complete.

The Naive Operator-Substitution and Its Limitations

At a superficial level, (D) is solved by setting

$$\varphi_n(x) = p_n\left(\frac{d}{dx}\right)\varphi_0(x),$$

where $\{p_n\}$ are orthogonal polynomials with recurrence coefficients $\{\alpha_n\}, \{\beta_n\}, \{\gamma_n\}$. This substitution works because applying the three-term recurrence to $\frac{d}{dx}$ reproduces exactly the structure of (D). However, this formulation has three deficiencies:

- It does not help design the differentiation matrix to be skew-symmetric unless non-positive measures μ are admitted.
- It does not elucidate the function space spanned by Φ .
- It does not determine when Φ is orthogonal.

The resolution of all three comes through Fourier analysis.

Prerequisites from Fourier Analysis

Fourier transform convention. For $f \in L_2(\mathbb{R})$,

$$\hat{f}(\xi) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-ix\xi} f(x) dx,$$

with the inverse $f(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{ix\xi} \hat{f}(\xi) d\xi$.

Parseval's identity. For $f, h \in L_2(\mathbb{R})$,

$$\int_{-\infty}^{\infty} f(x) \overline{h(x)} dx = \int_{-\infty}^{\infty} \hat{f}(\xi) \overline{\hat{h}(\xi)} d\xi.$$

Differentiation in the Fourier domain. Differentiation $\varphi'(x)$ corresponds to multiplication by $i\xi$ in the frequency domain: $\widehat{\varphi'}(\xi) = i\xi \hat{\varphi}(\xi)$. This is the key observation: the differential recurrence (D) in the spatial domain becomes a multiplicative recurrence in the frequency domain, and multiplicative recurrences are exactly what Favard's theorem governs.

Paley–Wiener space. For $\Omega \subseteq \mathbb{R}$, $\text{PW}_\Omega(\mathbb{R})$ is the space of all $f \in L_2(\mathbb{R})$ whose Fourier transform is supported on Ω .

Orthogonal polynomials on the real line. A probability measure $d\mu$ on $\mathbb{R}$ with all moments finite and infinitely many points of increase determines a unique orthonormal polynomial system $P = \{p_n\}_{n \in \mathbb{Z}_+}$ satisfying

$$\xi p_n(\xi) = b_{n-1} p_{n-1}(\xi) + c_n p_n(\xi) + b_n p_{n+1}(\xi),$$

with $b_n > 0$ and $c_n \in \mathbb{R}$ (after normalization). When $d\mu$ is symmetric, $c_n = 0$ for all n .

Theorem 1: The Fourier Characterization

Statement. A set of functions $\Phi = \{\varphi_n\}_{n \in \mathbb{Z}_+} \subset L_2(\mathbb{R})$ satisfies

$$\varphi'_0 = ic_0 \varphi_0 + b_0 \varphi_1,$$

$$\varphi'_n = -\overline{b_{n-1}} \varphi_{n-1} + ic_n \varphi_n + b_n \varphi_{n+1}, \quad n \geq 1,$$

for some sequences $b_n \in \mathbb{C} \setminus \{0\}$, $c_n \in \mathbb{R}$, if and only if

$$\varphi_n(x) = \frac{e^{i\theta_n}}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{ix\xi} p_n(\xi) g(\xi) d\xi, \quad (4)$$

where:

- $P = \{p_n\}_{n \in \mathbb{Z}_+}$ is an orthonormal polynomial system with respect to $d\mu$, a probability measure on $\mathbb{R}$ with all moments finite and infinitely many points of increase.
- $g \in L_2(\mathbb{R})$ satisfies $|\xi^n g(\xi)| \rightarrow 0$ as $|\xi| \rightarrow \infty$ for all $n \geq 0$.
- $\{\theta_n\}_{n \in \mathbb{Z}_+} \subset [0, 2\pi)$.

Why this works. In the Fourier domain, $\varphi'_n(x)$ corresponds to multiplication by $i\xi$. Writing $\hat{\varphi}_n(\xi) = \frac{e^{i\theta_n}}{\sqrt{2\pi}} p_n(\xi) g(\xi)$ (up to the Fourier convention sign), the differential recurrence becomes

$$i\xi \cdot p_n(\xi) g(\xi) = (\text{three-term recurrence in } p_n),$$

which is precisely the statement that $\{p_n\}$ are orthogonal polynomials for $d\mu$ with the recurrence $\xi p_n(\xi) = b_{n-1} p_{n-1}(\xi) + c_n p_n(\xi) + b_n p_{n+1}(\xi)$. The function g plays the role of a "spectral window"—it modulates the Fourier integral but does not participate in the recurrence structure.

The decay condition $|\xi^n g(\xi)| \rightarrow 0$ ensures that the integrals defining φ_n converge and that term-by-term differentiation under the integral sign is justified.

Corollary 1: Real-Valued Skew-Symmetric Case

Statement. A set of real-valued functions $\Phi = \{\varphi_n\}_{n \in \mathbb{Z}_+} \subset L_2(\mathbb{R})$ satisfies the purely skew-symmetric recurrence

$$\varphi'_0 = b_0 \varphi_1, \quad \varphi'_n = -b_{n-1} \varphi_{n-1} + b_n \varphi_{n+1}, \quad n \geq 1,$$

for some sequence $b_n > 0$, if and only if Φ satisfies the requirements of Theorem 1 with three additional constraints:

1. $d\mu$ is symmetric: $d\mu(-\xi) = d\mu(\xi)$.
2. g has even real part and odd imaginary part.
3. $\theta_n = n\pi/2 \pmod{2\pi}$, i.e. $e^{i\theta_n} = i^n$.

Role of each constraint. Symmetry of μ forces $c_n = 0$, eliminating the diagonal term. The parity condition on g and the phase i^n together ensure that φ_n is real-valued: the Fourier integral of $i^n p_n(\xi) g(\xi)$ with p_n having definite parity $(-1)^n$ under $\xi \mapsto -\xi$ (from the symmetry of μ) and g having the stated parity produces a real function.

The skew-symmetry $\gamma_n = -\alpha_{n+1}$ in the differentiation matrix (i.e. $b_n = b_n$ with $\alpha_n = -b_{n-1}$, $\gamma_n = b_n$) emulates the skew-self-adjointness of $\frac{d}{dx}$ on $L_2(\mathbb{R})$ with respect to zero boundary conditions at infinity.

Theorem 2: Orthogonality of the Basis

Statement. Let $\Phi = \{\varphi_n\}$ satisfy the requirements of Theorem 1. Then Φ is orthogonal in $L_2(\mathbb{R})$ if and only if P is orthogonal with respect to the measure $|g(\xi)|^2 d\xi$. Furthermore, whenever Φ is orthogonal, the functions $\varphi_n/\|g\|_2$ are orthonormal.

Proof. By Parseval's identity,

$$\int_{-\infty}^{\infty} \overline{\varphi_n(x)} \varphi_m(x) dx = e^{i(\theta_m - \theta_n)} \int_{-\infty}^{\infty} p_m(\xi) p_n(\xi) |g(\xi)|^2 d\xi.$$

The left side vanishes for all $n \neq m$ if and only if the right side does, which is precisely the statement that $\{p_n\}$ are orthogonal polynomials for $|g(\xi)|^2 d\xi$. When this holds, the squared norm of φ_n is $\|g\|_2^2$ (since p_n is orthonormal with respect to $d\mu = |g|^2 d\xi$), hence $\varphi_n/\|g\|_2$ is orthonormal.

The critical point. Theorem 1 requires orthogonality of P with respect to some probability measure $d\mu$. Theorem 2 adds that Φ is orthogonal in $L_2(\mathbb{R})$ precisely when $d\mu = |g(\xi)|^2 d\xi$ (up to normalization). This ties the spectral window g directly to the orthogonality measure.

Theorem 3: Completeness and Paley–Wiener Spaces

Statement. Let $\Phi = \{\varphi_n\}$ satisfy the requirements of Theorem 2 with a measure $d\mu$ such that polynomials are dense in $L_2(\mathbb{R}; d\mu)$. Then Φ forms an orthogonal basis for the Paley–Wiener space $PW_\Omega(\mathbb{R})$, where $\Omega = \text{supp}(d\mu)$.

Consequence for completeness in $L_2(\mathbb{R})$. The space $PW_\Omega(\mathbb{R})$ equals $L_2(\mathbb{R})$ if and only if $\Omega = \mathbb{R}$. Therefore, for Φ to be complete in $L_2(\mathbb{R})$, it is necessary that $d\mu$ be supported on all of $\mathbb{R}$. This immediately rules out Jacobi-seeded systems (supported on $(-1, 1)$) from being complete in $L_2(\mathbb{R})$, and explains why Hermite, (generalized) Laguerre, and continuous Hahn measures — all supported on $\mathbb{R}$ or half-lines that can be combined to cover $\mathbb{R}$ — are the natural candidates.

The density condition. The hypothesis that polynomials be dense in $L_2(\mathbb{R}; d\mu)$ is equivalent to $d\mu$ being a determinate moment problem (for compactly supported measures this is automatic; for measures on $\mathbb{R}$ it is a non-trivial condition related to Carleman's criterion).

The Canonical Form

For an absolutely continuous measure $d\mu(\xi) = w(\xi) d\xi$ with orthonormal polynomials $P = \{p_n\}$, the canonical associated orthonormal basis functions are:

$$\varphi_n(x) = \frac{i^n}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{ix\xi} p_n(\xi) |w(\xi)|^{1/2} d\xi. \quad (5)$$

This corresponds to the choices $\theta_n = n\pi/2$ and $g(\xi) = |w(\xi)|^{1/2}$ in Theorem 1. With this representation, all coefficients b_n are positive real, and the differentiation matrix entries correspond directly to the three-term recurrence coefficients:

$$\begin{aligned}\xi p_n(\xi) &= b_{n-1}p_{n-1}(\xi) + c_n p_n(\xi) + b_n p_{n+1}(\xi) \\ \iff \varphi'_n(x) &= -b_{n-1}\varphi_{n-1}(x) + ic_n \varphi_n(x) + b_n \varphi_{n+1}(x).\end{aligned}$$

The Differentiation Matrix

In a spectral method with basis Φ , the derivative of $u_N(x) = \sum_{n=0}^N a_n \varphi_n(x)$ is $u'_N(x) = \sum_{n=0}^N \tilde{a}_n \varphi_n(x)$ where $\tilde{\mathbf{a}} = D\mathbf{a}$ and

$$D = \begin{pmatrix} \beta_0 & \alpha_1 & & & \\ \gamma_0 & \beta_1 & \alpha_2 & & \\ & \gamma_1 & \beta_2 & \ddots & \\ & & \ddots & \ddots & \alpha_N \\ & & & \gamma_{N-1} & \beta_N \end{pmatrix}. \quad (3)$$

The desiderata for D are:

- **Skew-symmetry** ($\gamma_n = -\alpha_{n+1}$): emulates skew-self-adjointness of $\frac{d}{dx}$, yielding ℓ_2 -stability for time-dependent PDEs and norm-preservation for dispersive equations.
- **Orthonormality of Φ** : equates the discrete ℓ_2 norm on coefficients with the continuous L_2 norm of the numerical solution.
- **Completeness of Φ** : ensures all functions of interest are approximable.

The Worked Examples

Hermite Functions

The Hermite measure $d\mu(\xi) = e^{-\xi^2} d\xi$ is symmetric and supported on $\mathbb{R}$. The canonical transform (5) produces exactly the Hermite functions

$$\varphi_n(x) = \frac{(-1)^n}{(2^n n!)^{1/2} \pi^{1/4}} e^{-x^2/2} H_n(x), \quad n \in \mathbb{Z}_+,$$

with the skew-symmetric recurrence

$$\varphi'_n(x) = -\sqrt{\frac{n}{2}} \varphi_{n-1}(x) + \sqrt{\frac{n+1}{2}} \varphi_{n+1}(x).$$

Here $b_n = \sqrt{(n+1)/2}$ and $c_n = 0$. The Hermite functions are the unique system within this framework having the product form $\varphi_n(x) = h(x)q_n(x)$ with $h \in L_2(\mathbb{R})$ and q_n a polynomial of degree n . They satisfy the Cramér inequality $|\varphi_n(x)| \leq \pi^{-1/4}$ for all $x \in \mathbb{R}$, $n \in \mathbb{Z}_+$.

Since $d\mu$ is supported on all of $\mathbb{R}$ and the Hermite measure is determinate, Theorem 3 yields completeness: Φ is an orthonormal basis for $L_2(\mathbb{R})$.

Transformed Legendre Functions (Ultraspherical, $\alpha = 0$)

The Legendre measure $d\mu(\xi) = \chi_{(-1,1)}(\xi) d\xi$ gives the remarkably compact

$$\varphi_n(x) = \sqrt{\frac{n + \frac{1}{2}}{x}} J_{n+1/2}(x), \quad n \in \mathbb{Z}_+,$$

where J_ν is the Bessel function of order ν . The recurrence coefficients are

$b_n = \sqrt{(n+1)(n+1)/(2n+1)(2n+3)}$ (specializing the ultraspherical formula with $\alpha = 0$) and $c_n = 0$. These live in $\text{PW}_{(-1,1)}(\mathbb{R})$ — they are bandlimited functions — hence they are not complete in $L_2(\mathbb{R})$ by Theorem 3.

More generally, for the ultraspherical measure $d\mu(\xi) = \chi_{(-1,1)}(\xi)(1 - \xi^2)^\alpha d\xi$ with $\alpha > -1$, the first two transformed functions involve Bessel functions:

$$\varphi_0(x) = C_\alpha \left(\frac{2}{x}\right)^{(1+\alpha)/2} J_{(1+\alpha)/2}(x), \quad \varphi_1(x) = C_\alpha \sqrt{3+2\alpha} \left(\frac{2}{x}\right)^{(1+\alpha)/2} J_{(3+\alpha)/2}(x),$$

with $C_\alpha^2 = \Gamma(\alpha + 3/2)\Gamma(\alpha/2 + 1)/(2^{\alpha+1}\Gamma((1 + \alpha)/2))$.

Malmquist–Takenaka Functions (from Laguerre)

The Laguerre measure $d\mu(\xi) = \chi_{[0,\infty)}(\xi)e^{-\xi} d\xi$ produces

$$\varphi_n(x) = \sqrt{\frac{2}{\pi}} i^n \frac{(1 + 2ix)^n}{(1 - 2ix)^{n+1}}, \quad n \in \mathbb{Z}_+.$$

These are dense in $\text{PW}_{[0,\infty)}(\mathbb{R})$, not in $L_2(\mathbb{R})$. Completeness in $L_2(\mathbb{R})$ is obtained by taking the direct sum with the reflected Laguerre measure $d\mu(\xi) = \chi_{(-\infty,0]}(\xi)e^\xi d\xi$, yielding the Malmquist–Takenaka system indexed over $n \in \mathbb{Z}$:

$$\varphi_n(x) = \sqrt{\frac{2}{\pi}} i^n \frac{(1 + 2ix)^n}{(1 - 2ix)^{n+1}}, \quad n \in \mathbb{Z}. \quad (8)$$

Under the change of variables $\theta = 2 \arctan(2x)$,

$$\varphi_n(x) = \sqrt{\frac{2}{\pi}} i^n e^{i(n+1/2)\theta} \cos \frac{\theta}{2}. \quad (9)$$

This reveals the Malmquist–Takenaka basis as the Fourier basis in disguise: it is a system of complex exponentials $e^{i(n+1/2)\theta}$ modulated by $\cos(\theta/2)$, under a Möbius-type change of variable. This identification yields an $O(N \log N)$ FFT-based algorithm for computing expansion coefficients.

Tanh–Jacobi Functions (from Continuous Hahn)

The continuous Hahn measure

$$d\mu_{a,b}(\xi) = \frac{1}{2\pi} |\Gamma(a + i\xi)\Gamma(b - i\xi)|^2 d\xi, \quad \xi \in \mathbb{R}, \quad \operatorname{Re}(a), \operatorname{Re}(b) > 0,$$

is supported on all of $\mathbb{R}$, so the resulting system is complete in $L_2(\mathbb{R})$. The canonical transform produces the tanh–Jacobi functions:

$$\varphi_n^{a,b}(x) = (1 - \tanh x)^a (1 + \tanh x)^b p_n^{(2a-1, 2b-1)}(\tanh x), \quad n \in \mathbb{Z}_+, \quad (12)$$

where $p_n^{(\alpha, \beta)}$ is the n th Jacobi polynomial normalized with respect to $(1 - \xi)^\alpha (1 + \xi)^\beta$ on $(-1, 1)$. This identity generalizes a result of Ramanujan:

$$\int_{-\infty}^{\infty} |\Gamma(a + i\xi)|^2 e^{ix\xi} d\xi = \frac{\sqrt{\pi} \Gamma(a) \Gamma(a + \frac{1}{2})}{\cosh^{2a}(x/2)}, \quad a > 0.$$

When $(a, b) \in \{(\frac{1}{4}, \frac{1}{4}), (\frac{1}{4}, \frac{3}{4}), (\frac{3}{4}, \frac{1}{4}), (\frac{3}{4}, \frac{3}{4})\}$, the Jacobi polynomials become Chebyshev polynomials of one of four kinds, and expansion coefficients are computable via Fast Cosine Transform in $O(N \log N)$ operations.

Generalized Laguerre and Polynomials on the Unit Circle

The generalized Laguerre measure $d\mu(\xi) = \chi_{(0, \infty)}(\xi) \xi^\alpha e^{-\xi} d\xi$ with $\alpha > -1$ gives

$$\varphi_n(x) = (-i)^n \sqrt{\frac{2}{\pi}} \left(\frac{1}{1 - 2ix} \right)^{1+\alpha/2} \Pi_n^{(\alpha)} \left(\frac{1 + 2ix}{1 - 2ix} \right),$$

where $\Pi_n^{(\alpha)}$ is a polynomial of degree n . Under $x = \frac{1}{2} \tan(\theta/2)$, i.e.

$(1 + 2ix)/(1 - 2ix) = e^{i\theta}$, these polynomials become orthogonal polynomials on the unit circle (OPUC) with weight $W(\theta) = \cos^\alpha(\theta/2)$, related to the Szegő–Askey polynomials $\phi_n^{(\lambda)}$ by $\Pi_n^{(\alpha)}(z) \propto \phi_n^{(\alpha/2)}(-z)$, and in turn to Jacobi polynomials $P_n^{(\frac{\alpha-1}{2}, -\frac{1}{2})}$ and $P_n^{(\frac{\alpha+1}{2}, \frac{1}{2})}$ via the Delsarte–Genin relationship.

The Non-Uniqueness of φ_0 and the Phase Freedom

A subtle distinction between the classical and differential settings: orthogonal monic polynomials are determined uniquely by the Jacobi matrix. In the differential analogue, the differentiation matrix D — i.e. the coefficients b_n and c_n — is insufficient to determine Φ . One must also specify φ_0 , yet not every choice of φ_0 yields an orthonormal system.

The canonical form (5) can be generalized to

$$\varphi_n(x) = \frac{i^n}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{ix\xi} p_n(\xi) e^{i\sigma(\xi)} |g(\xi)|^{1/2} d\xi, \quad (14)$$

for any measurable $\sigma : \text{supp}(\mu) \rightarrow \mathbb{R}$. All such bases are orthonormal in $L_2(\mathbb{R})$ by Theorem 2, and all share exactly the same differentiation matrix: b_n and c_n do not depend on σ . The phase function σ captures the additional degree of freedom in specifying φ_0 .

This freedom has a direct physical consequence for the Schrödinger equation $\partial u / \partial t = i \partial^2 u / \partial x^2 + i F(x, u)$. Each φ_n evolves under the free equation $\partial u / \partial t = i \partial^2 u / \partial x^2$ to

$$\psi_n(x, t) = \frac{i^n}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{ix\xi} p_n(\xi) e^{i\xi t^2} |g(\xi)|^{1/2} d\xi,$$

which amounts to replacing $\sigma(\xi)$ by $\sigma(\xi) + \xi t^2$. The basis "rotates" in phase space under free Schrödinger evolution.

Periodic Bases from Discrete Orthogonal Polynomials

The framework extends to $L_2(-\pi, \pi)$. Let $\mathcal{Z} \subseteq \mathbb{Z}$ have infinite cardinality, and let $\{\sigma_k\}_{k \in \mathcal{Z}}$ be positive weights normalized so that $\sum_k \sigma_k = 1$. The discrete inner product $\langle f, h \rangle = \sum_{k \in \mathcal{Z}} \sigma_k \overline{f(k)} h(k)$ determines orthonormal polynomials $\{p_n\}$. The 2π -periodic functions

$$\varphi_n(x) = i^n \sum_{k \in \mathcal{Z}} \sqrt{\sigma_k} p_n(k) e^{ikx}, \quad n \in \mathbb{Z}_+,$$

satisfy $\varphi'_n(x) = -b_{n-1} \varphi_{n-1}(x) + c_n i \varphi_n(x) + b_n \varphi_{n+1}(x)$ and are orthonormal on $L_2(-\pi, \pi)$ by Parseval's identity for Fourier series:

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} \overline{\varphi_m(x)} \varphi_n(x) dx = i^{m-n} \sum_{k \in \mathcal{Z}} \sigma_k p_m(k) p_n(k) = \delta_{m,n}.$$

Theorems 1–3 generalize naturally to this periodic setting.

Approximation Theory and Convergence Rates

The rate of decay of expansion coefficients $\hat{f}_n$ in a basis Φ determines the convergence speed of partial sums. For the Malmquist–Takenaka basis, the following rates illustrate a striking non-uniformity:

		Character
$1/(1+x^4)$	$O(\rho^{-\ n\ }), \rho = 1 + \sqrt{2}$	Exponential (geometric)
$\sin x/(1+x^4)$	$O(\ n\ ^{-9/4})$	Algebraic
$\sin x/(1+x^2)$	$O(\ n\ ^{-5/4})$	Algebraic
e^{-x^2}	$O(e^{-3\ n\ ^{2/3}/2})$	Subexponential

$f(x)$	$\hat{f}_n$	Character
$1/\cosh x$	$O(e^{-2\ n\ ^{1/2}})$	Subexponential

The collapse from exponential to algebraic decay when $\sin x$ appears is caused by $\sin x$ having an essential singularity at ∞ on the Riemann sphere. The classical Bernstein-ellipse theory, which governs approximation on compact intervals, fails entirely on $\mathbb{R}$, and no overarching replacement theory exists. Each function requires individual analysis via steepest descent.

The Growth of Recurrence Coefficients and Freud Polynomials

For spectral methods, the spectral radius of the truncated differentiation matrix D_N controls the stiffness of time-stepping: smaller off-diagonal entries b_n allow larger time steps τ . The connection to orthogonal polynomial recurrence coefficients implies $\limsup_{n \rightarrow \infty} b_n = +\infty$ (this is an elementary consequence of the theory: any measure supported on an unbounded set has unbounded recurrence coefficients). For Hermite, $b_n = \sqrt{(n+1)/2} = O(n^{1/2})$; for Malmquist-Takenaka, $b_n = n+1 = O(n)$.

Freud polynomials, orthogonal with respect to $e^{-|x|^\sigma}$, $\sigma > 0$, offer the growth rate $b_n = O(n^{1/\sigma})$ by the Freud conjecture (proved by Lubinsky, Mhaskar, and Saff). Larger σ gives slower growth. However, except for Hermite ($\sigma = 2$), the explicit form of Freud polynomials and their recurrence coefficients are unknown, and the string relations governing these coefficients are numerically unstable. This is the paper's call for more polynomials: explicitly known orthogonal polynomial families on $\mathbb{R}$ with slowly growing recurrence coefficients.

Summary of Transform Pairs

Polynomials p_n	Special functions associated to φ_n
Hermite	Hermite functions/polynomials
Laguerre	Malmquist-Takenaka functions, Fourier basis, Chebyshev polynomials
Generalized Laguerre	Szegő-Askey polynomials on the unit circle, Jacobi polynomials
Ultraspherical	Bessel functions
Continuous Hahn	Jacobi polynomials

Recapitulation

The differential analogue of Favard's theorem characterizes sequences $\Phi = \{\varphi_n\}_{n \in \mathbb{Z}_+} \subset \mathbf{L}_2(\mathbb{R})$ whose derivatives satisfy a three-term recurrence $\varphi'_n = \alpha_n \varphi_{n-1} + \beta_n \varphi_n + \gamma_n \varphi_{n+1}$. The characterization proceeds by passing to the Fourier domain, where differentiation becomes multiplication, reducing the problem to the classical

Favard setting. Each such system has the integral representation

$\varphi_n(x) = \frac{e^{i\theta_n}}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{ix\xi} p_n(\xi) g(\xi) \, d\xi$, with $\{p_n\}$ orthogonal polynomials for a measure $d\mu$ and g a spectral window. Orthogonality of Φ in $L_2(\mathbb{R})$ holds precisely when $d\mu = |g|^2 d\xi$. Completeness of Φ in $L_2(\mathbb{R})$ requires $\text{supp}(d\mu) = \mathbb{R}$; otherwise Φ spans a Paley–Wiener space $\mathbf{PW}_\Omega(\mathbb{R})$. The differentiation matrix D is tridiagonal with entries inherited from the polynomial recurrence, and is skew-symmetric when $d\mu$ is symmetric. Unlike the classical setting, the differentiation matrix does not uniquely determine the basis — the additional freedom resides in the choice of φ_0 (equivalently, a phase function σ in the Fourier integral), with direct consequences for evolution PDEs.